

Model engineering and river engineering exercises at the Oberrach Research Station



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1.0 MODEL 1: MOOSBURG TEST MODEL

1.1. INTRODUCTION AND SCOPE OF THE MODEL

Hydraulic structures such as weirs play a crucial role in water resource management, flow regulation, and flood control. The performance and safety of such systems depend on a precise understanding of flow behavior under varying operational and environmental conditions. In this context, physical modeling remains an essential tool for investigating complex hydraulic processes that are difficult to capture analytically.

This study investigates a physical model of the *Moosburg* weir system, developed at a geometric scale of 1:25. The model is designed according to Froude similarity principles to maintain dynamic similarity between model and prototype conditions. As a result, key free-surface flow processes such as discharge distribution, energy dissipation, and overflow characteristics can be reliably reproduced.



Figure 1: Moosburg River Aerial View.

The primary objective of this investigation is to analyze the hydraulic performance of the weir system under different operating conditions. This includes the assessment of regular operation with distributed discharge through the channel intake and fish passes, as well as the evaluation of extreme conditions where only the weir bay with a radial gate is active.



Figure 2: Moosburg River Model (experimental setup).

A key aspect of the study is the regulation of the upstream water level at a defined target elevation. This involves examining the relationship between discharge, gate operation, and resulting water levels. Additionally, the capacity of the weir bay is determined by analyzing discharge–water level relationships and calculating the discharge coefficient based on the Poleni equation.

Another important component of the investigation is the influence of floating debris on the hydraulic performance. The accumulation of wooden debris at the intake structure is studied to evaluate its impact on flow conditions, potential blockage formation, and associated risks such as reduced discharge capacity and upstream flooding. The effectiveness of operational measures, such as flushing using the radial gate, is also assessed.

1.2 EXPERIMENTAL SETUP

The physical model represents an integrated hydraulic system consisting of a weir bay equipped with a radial gate, an intake channel supplying a hydropower plant, and fish passages located both upstream and downstream. The upstream section includes a floodplain area, where the flow approaches the weir under controlled discharge conditions. The overflow across the weir is regulated by the radial gate, which controls the upstream water level. At the same time, a portion of the discharge is diverted through the intake channel towards the powerhouse, while the remaining flow passes over the weir crest. To ensure

ecological continuity, fish passages are provided on both sides of the structure: one facilitates downstream migration, and the other supports upstream movement.

Water levels were measured at designated gauge locations. The upstream water levels were recorded at OWL2 (left side) and OWR3 (right side), while the downstream conditions were monitored using a tailwater gauge. These measurements were used to evaluate the stage–discharge relationship and to analyze the hydraulic behavior of the system. The discharge was controlled using a laboratory flow regulation system, allowing precise adjustment of flow conditions. All measurements were conducted under steady-state conditions, ensuring reliable and consistent results.

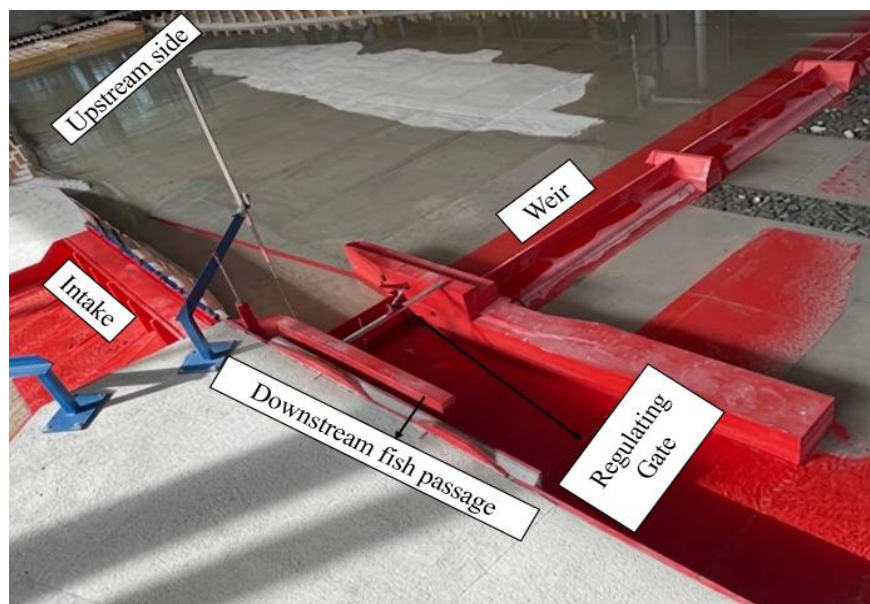


Figure 3: Experimental Setup.

1.3 TASK 1

Under regular operating conditions, the total discharge of $90 \text{ m}^3/\text{s}$ is distributed between the channel intake and the fish passages. A discharge of $70 \text{ m}^3/\text{s}$ is diverted through the intake channel towards the powerhouse, while $10 \text{ m}^3/\text{s}$ is conveyed through each of the upstream and downstream fish passages. This distribution ensures both hydraulic functionality and ecological continuity. The system is operated under steady-state conditions, allowing the establishment of stable flow patterns. The flow approaching the weir and intake structure is generally uniform, with no significant disturbances observed under normal conditions. The majority of the discharge is efficiently conveyed through the intake channel, while the remaining flow is safely guided through the fish passages.

Task 1 was designed to investigate the hydraulic behavior of the system under both regular operating conditions and debris-influenced conditions in a controlled laboratory environment. The discharge was first set to $90 \text{ m}^3/\text{s}$ using the laboratory flow control system. Subsequently, the upstream water level was adjusted and maintained at approximately 412.43 m asl, corresponding to 39.72 cm on the gauge, by regulating the opening of the radial gate. Once the desired conditions were achieved, the system was allowed to stabilize to ensure steady-state flow, providing reliable conditions for observation and analysis. For Task 1.1, the experiment was carried out under clean flow conditions, meaning that no obstructions were present near the intake structure. The flow was distributed through the intake channel, regulating gate, and both fish passages. Observations focused on the overall flow behaviour, including the uniformity of the approach flow, the efficiency of flow conveyance into the intake, and the presence of turbulence or flow separation. Particular attention was given to ensuring that the flow remained stable and evenly distributed, indicating proper hydraulic performance of the system under normal operation.

For Task 1.2, wooden debris was introduced near the intake channel to simulate the accumulation of driftwood commonly observed in natural river systems. To ensure a meaningful comparison with the regular operation case, the same discharge and boundary conditions were maintained. The experiment aimed to assess how the presence of debris alters the flow conditions. Observations focused on changes in flow distribution, the development of turbulence, the formation of recirculation zones, and the resulting impact on the hydraulic efficiency of the intake. In addition, the potential reduction in effective flow area and its influence on upstream water levels were considered.

1.3.1 OBSERVATION AND RESULT

1.3.1.1 Under regular operating conditions

The upstream flow was observed to be subcritical, with a stable water surface profile and relatively low velocity gradients. The absence of noticeable turbulence indicates that the flow regime is hydraulically stable, with minimal energy dissipation in the upstream reach. The discharge was evenly distributed between the intake channel, fish passages, and downstream section. No significant flow separation was detected; only minor localized recirculation zones occurred near structural elements, which did not affect the overall flow behaviour (Figure 4). The approach flow towards the intake channel remained well-structured and streamlined, ensuring efficient hydraulic performance of the intake system.



Figure 4 Ink test at upstream of Power Plant.

In the fish passages, the upstream passage exhibited low velocities and low turbulence intensity, with no observable recirculation zones, providing favourable conditions for fish migration. As stated in (Figure 5 & 6), the flow remains uniform and undisturbed within this section. The downstream fish passage showed slightly higher velocities; however, the flow remained stable, without the formation of vortices or flow instabilities.



Figure 5 Ink test at upstream of Fish pass.



Figure 6 Ink test at downstream of Fish Pass.

Task 1.2-Test With Debris

During the experiment, wooden debris of different sizes and shapes was introduced near the intake region to simulate realistic driftwood conditions. The debris remained floating on the water surface and was transported along the flow path due to sufficient approach velocity in the upstream region. No significant accumulation of debris was observed at the intake entrance, and therefore no blockage formation occurred. The flow approaching the intake remained relatively uniform, and the overall flow pattern was not significantly altered (Figure 7). Only minor, localized disturbances were observed around individual debris elements, which did not develop into larger turbulence structures or recirculation zone. The debris did not enter the intake channel, indicating that the intake geometry and prevailing flow conditions effectively prevented debris intrusion. As a result, the effective flow area of the intake was not reduced, and no increase in hydraulic resistance was observed. This is further confirmed by the upstream water level, which remained essentially constant throughout the experiment, showing no signs of backwater effects.

Overall, the system maintained stable hydraulic conditions, with consistent flow distribution between the intake, fish passages, and downstream section. The results indicate that under the given operating conditions, the system is capable of handling floating debris without any significant impact on hydraulic performance.

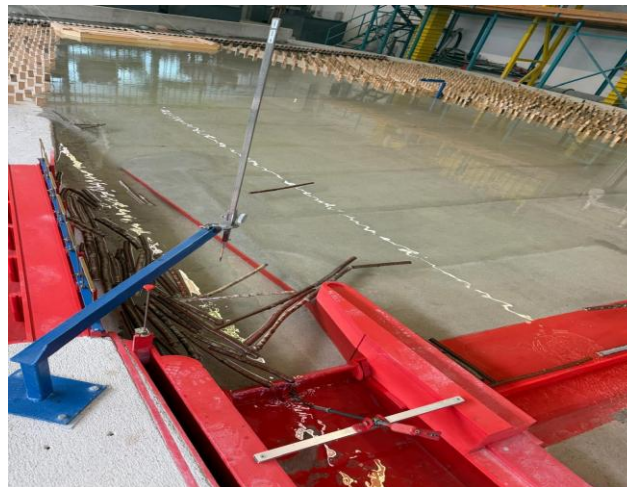


Figure 7 Debris Jam.

Result

The results obtained from Task 1 demonstrate the influence of operating conditions on the hydraulic behaviour of the system. Under regular operating conditions, the flow remains subcritical and stable, indicating well-balanced hydraulic performance. The smooth water surface profile and uniform flow distribution suggest that velocities are appropriately controlled, and energy losses occur gradually without causing disturbances. This results in an efficient and well-aligned approach flow toward the intake structure.

In addition, when debris is introduced near the intake, only minor changes in the flow pattern were observed. The debris remained floating and did not significantly reduce the effective flow area. As a result, no major increase in velocity or strong turbulence was detected. The flow largely maintained its original path, with only small, localized disturbances forming around individual debris elements. No significant flow separation or large recirculation zones were observed, indicating that the system remained hydraulically stable even in the presence of debris. The overall flow distribution toward the intake and fish passages remained relatively uniform, and the intake performance was not noticeably affected. Similarly, the fish passages continued to operate effectively, with only minor variations in local flow conditions.

The absence of blockage and stable upstream water levels indicate that the system is capable of handling floating debris without introducing additional hydraulic resistance or energy losses. Once the debris was flushed out, the flow conditions returned to their original state, confirming the robustness of the system under the tested conditions.

1.4 TASK 2

The experimental conditions for Task 2 were defined to evaluate the hydraulic capacity of the weir bay under controlled boundary conditions. In this configuration, the intake channel and both fish passages were closed, ensuring that the entire discharge was conveyed exclusively through the weir bay controlled by the radial gate. This arrangement isolates the weir and allows a direct assessment of its discharge capacity without interference from other flow paths.

The discharge was varied over a range from 70 m³/s to 170 m³/s using the laboratory flow control system. For each discharge setting, sufficient time was allowed for the flow to reach steady-state conditions before any observations or measurements were recorded. This ensured that the results represent stable hydraulic behaviour. Water levels were measured at the upstream gauge locations, namely OWL2 and OWR3, for each discharge condition. These measurements were used to determine the hydraulic head over the weir crest, which represents the driving force for the overflow. The head was calculated using the following relationship:

$$H = h - z_{crest}$$

where H is the head over the crest, h is the measured upstream water level, and $z_{crest} = 408.00$ m asl is the crest elevation of the weir.

The discharge over the weir was analyzed using the Poleni equation, which describes free overflow conditions. To evaluate the hydraulic performance of the weir, the equation was rearranged to determine the discharge coefficient:

$$\mu = \frac{Q}{b \cdot \sqrt{2g} \cdot H^{3/2}}$$

This coefficient represents the efficiency of the weir and accounts for energy losses. In addition to the recorded measurements, qualitative observations were carried out for each discharge condition. These focused on variations in flow velocity, the formation and location of the hydraulic jump, as well as changes in turbulence and flow behaviour downstream of the weir.

1.4.1 OBSERVATIONS AND RESULTS

The flow behaviour was assessed by varying the discharge within a range of $70 \text{ m}^3/\text{s}$ to $170 \text{ m}^3/\text{s}$ in increments of $20 \text{ m}^3/\text{s}$. The closure of the intake channel and fish passages ensured that the entire discharge was routed exclusively over the weir through the radial gate. This controlled configuration enabled a detailed evaluation of the overflow characteristics and hydraulic performance of the structure.

For lower discharges ($70\text{--}90 \text{ m}^3/\text{s}$), the flow exhibited stable subcritical conditions ($Fr < 1$). The water surface remained smooth, and the flow pattern was uniform with minimal turbulence.



Figure 8 Flow condition at $70 \text{ m}^3/\text{s}$.

No hydraulic jump was observed in this range, indicating low energy dissipation and stable flow conditions.



Figure 9 Flow condition at 90 m³/s.

At a discharge of 110 m³/s, the flow began to transition. A weak and unstable hydraulic jump started to develop near the regulating gate (Figure 10). This was accompanied by a slight increase in turbulence and localized disturbances, particularly in regions influenced by structural elements. The flow was approaching critical conditions, marking the onset of a regime change.



Figure 10 Flow condition at 110 m³/s.

With further increase in discharge to $130 \text{ m}^3/\text{s}$, the hydraulic jump became more pronounced and shifted upstream. The flow exhibited clearer signs of supercritical to subcritical transition, with increased turbulence intensity and the formation of localized flow separation zones. The water surface became irregular, reflecting higher energy levels and increased flow instability (Figure 11).



Figure 11 Flow condition at $130 \text{ m}^3/\text{s}$.

At higher discharges ($150\text{--}170 \text{ m}^3/\text{s}$), the flow was characterized by a fully developed hydraulic jump, accompanied by strong turbulence, eddies, and recirculation zones. The jump moved further upstream, approaching the regulating gate. Upstream of the jump, the flow was supercritical ($Fr > 1$), while downstream conditions returned to subcritical flow ($Fr < 1$).



Figure 12 Flow condition at 150 m³/s.

There is also a contraction effect due to the pier is visible. Overall, it was observed that increasing discharge leads to a clear transition from stable, subcritical flow conditions to more complex flow behaviour dominated by hydraulic jump formation, turbulence, and energy dissipation.



Figure 13 Flow condition at 170 m³/s.

1.4.2 ANALYSIS OF RESULT

The results obtained from Task 2 demonstrate a clear dependence of the hydraulic behaviour on the discharge, particularly in terms of water level variation, flow regime transition, and energy dissipation. The measured water levels at OWL2 and OWR3 generally increase with discharge, indicating a rise in hydraulic head, although minor deviations occur due to local flow conditions.

Q (m ³ /s)	TWL (m)	OWL2 (m)	OWR3 (m)
70	410.15	410.76	410.49
90	410.40	410.975	410.725
110	410.62	410.862	410.49
130	410.82	412.17	411.375
150	411.02	411.95	411.65
170	411.19	412.2	411.85

Table 1 Calculated Water level.

At lower discharge values (70–90 m³/s), the flow remains predominantly subcritical ($Fr < 1$), characterized by relatively high flow depth and low velocity. The measured water levels increase from 410.76 m to 410.975 m at OWL2 and from 410.49 m to 410.725 m at OWR3, showing a gradual and consistent rise. The flow in this range is stable, with a smooth water surface and no visible hydraulic jump. Turbulence levels are low, and the energy is dissipated gradually without significant disturbances. At a discharge of 110 m³/s, a noticeable deviation is observed in the measured values, with OWL2 recorded at 410.86 m and OWR3 at 410.49 m. Compared to the previous values, this indicates a temporary reduction in upstream water level at OWL2. This behaviour suggests localized flow effects near the regulating gate, such as flow acceleration, redistribution, or minor instabilities. At this stage, the flow begins to show transitional characteristics, with small disturbances and the initial formation of unsteady flow patterns.

As the discharge increases to 130 m³/s, a significant rise in water levels is observed, with OWL2 reaching 412.17 m and OWR3 411.375 m. This indicates a substantial increase in hydraulic head and energy within the system. In this range, a hydraulic jump becomes clearly visible, marking the transition from supercritical flow upstream of the jump (high velocity and shallow depth) to subcritical flow

downstream (lower velocity and greater depth). The hydraulic jump plays an important role in energy dissipation, converting kinetic energy into turbulence and mixing.

At higher discharge values (150–170 m³/s), the water levels continue to increase, with OWL2 reaching 411.95 m and 412.20 m, and OWR3 reaching 411.65 m and 411.85 m, respectively. The hydraulic jump becomes more pronounced and shifts further upstream toward the regulating gate. This upstream movement is associated with increasing water levels and changing flow conditions, which force the jump to adjust its position. The flow in this range is highly energetic, with strong turbulence, recirculation zones, and air entrainment, indicating significant energy losses. In addition, localized velocity increases due to contraction effects near structural elements contribute to the complexity of the flow pattern.

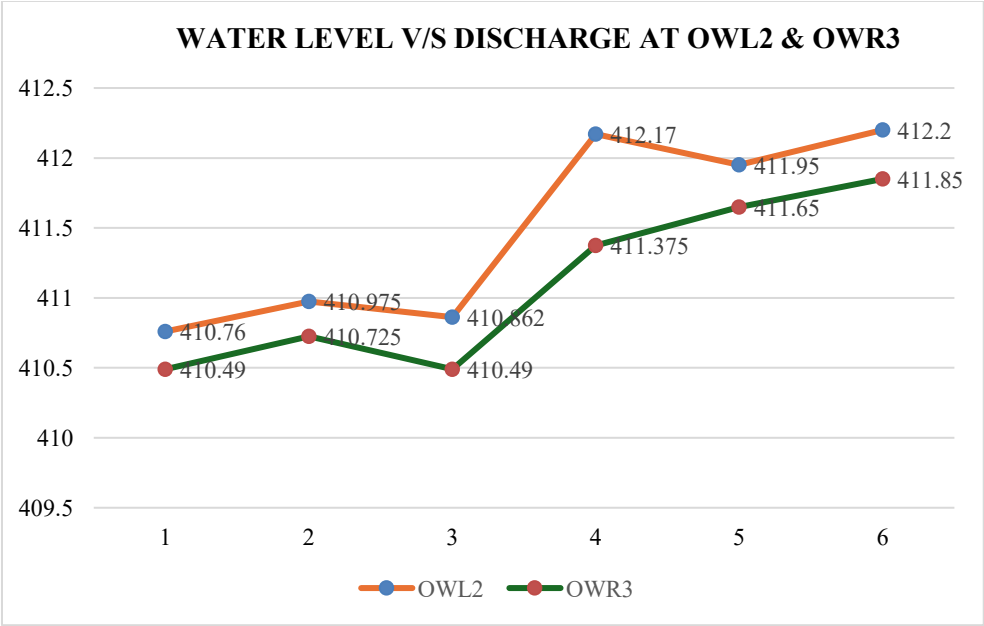


Figure 14 Stage discharge relationship at OWL2 and OWR3.

The stage–discharge relationship obtained from the measured data shows a generally nonlinear increase in upstream water level with increasing discharge, which is consistent with typical weir flow behavior. As the discharge increases from 70 m³/s to 170 m³/s, both OWL2 and OWR3 water levels increase, indicating a rise in hydraulic head. It can be observed from the graph that the water levels at OWL2 are consistently higher than those at OWR3, which suggests a slight asymmetry in the flow distribution across the channel width, possibly due to local flow conditions near the gate and structural geometry.

A small deviation in the trend is observed at $Q = 110 \text{ m}^3/\text{s}$, where a slight decrease in water level occurs compared to the previous discharge. This variation does not follow the expected hydraulic behavior and can be attributed to measurement error or minor experimental inconsistencies, rather than a physical change in flow conditions.

Q (m^3/s)	Poleni Shape Factor (μ)
70	0.413586
90	0.475164
110	0.675489
130	0.413591
150	0.517639
170	0.535065

Table 2 Calculated Poleni shape Factor.

The calculated Poleni coefficient (μ) ranges between 0.41 and 0.68, which falls within the expected range for weir flow. At lower discharges, μ values are relatively stable, while a peak at $110 \text{ m}^3/\text{s}$ and a drop at $130 \text{ m}^3/\text{s}$ indicate minor variations due to turbulence, flow transition, or measurement error.

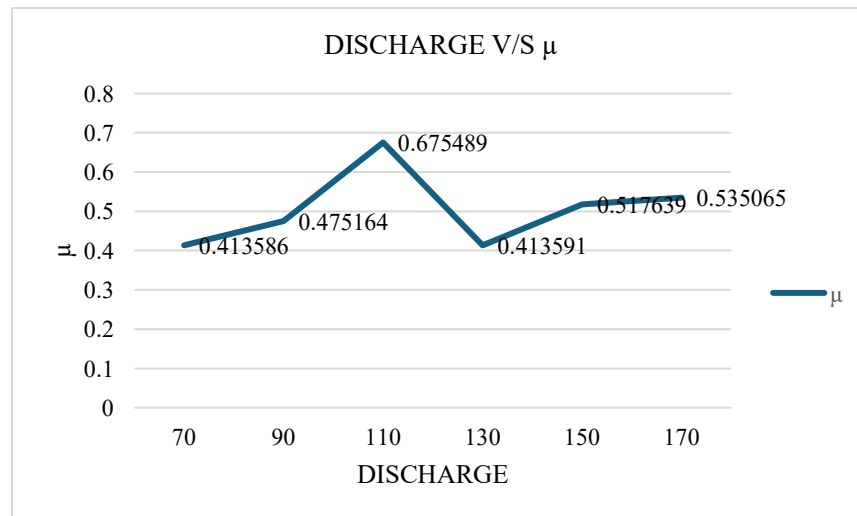


Figure 15 Variation of Poleni Coefficient with discharge.

1.5 CONCLUSION

The experimental investigation of the Moosburg weir model demonstrates that the hydraulic behavior of the system is significantly influenced by operating conditions and discharge. Under normal operating conditions, the flow remains subcritical, stable, and uniformly distributed, ensuring efficient performance of the intake structure.

The introduction of debris results in localized flow disturbances, including increased turbulence and minor recirculation effects. Although no major blockage was observed, these disturbances highlight the importance of effective debris management in maintaining optimal hydraulic efficiency. With increasing discharge, the flow exhibits a clear transition from stable conditions to more complex behavior, characterized by the formation of a hydraulic jump. This phenomenon plays a key role in energy dissipation and progressively shifts upstream with increasing discharge, reflecting the influence of tailwater conditions and momentum balance within the system.

The results indicate that the system maintains reliable hydraulic performance over a wide range of operating conditions, while remaining sensitive to variations in discharge and local flow disturbances. These findings emphasize the importance of appropriate design and operational strategies to ensure stable and efficient system performance

2.0 LANDQUART MODEL TEST

2.1 INTRODUCTION AND SCOPE OF INVESTIGATION

The objective of this experiment is to assess a physical scale model of the Landsquart River in order to investigate its hydraulic responsiveness and performance under specified boundary circumstances. Understanding the flow behaviour in the river channel and surrounding the intake structure is the main goal of the study since it is crucial for efficient hydraulic design and river management.

The model test highlights the integration of engineering and environmental considerations in the planning and design of river infrastructure. Through this experiment, important information is obtained regarding flow patterns, sediment transport, and sediment deposition under various discharge conditions. Furthermore, the Landsquart River's reduced-scale model provides an effective foundation for optimising the weir's geometry, the settling basin's size, and the fishway's functionality. This ensures that the hydropower project's operating needs are met while preserving the river system's equilibrium in nature and environmental sustainability.



Figure 16 Landquart River model.

2.2 EXPERIMENTAL SETUP

The experiment was performed using a 1:15 geometrically scaled physical model of the Landsquart River. The model represents the river intake section together with its associated hydraulic structures. As shown in the figure (17), the setup includes a block ramp, intake gate, intake channel, flushing gate, sand trap, sediment flush, fish pass, and the intake leading to the power plant.

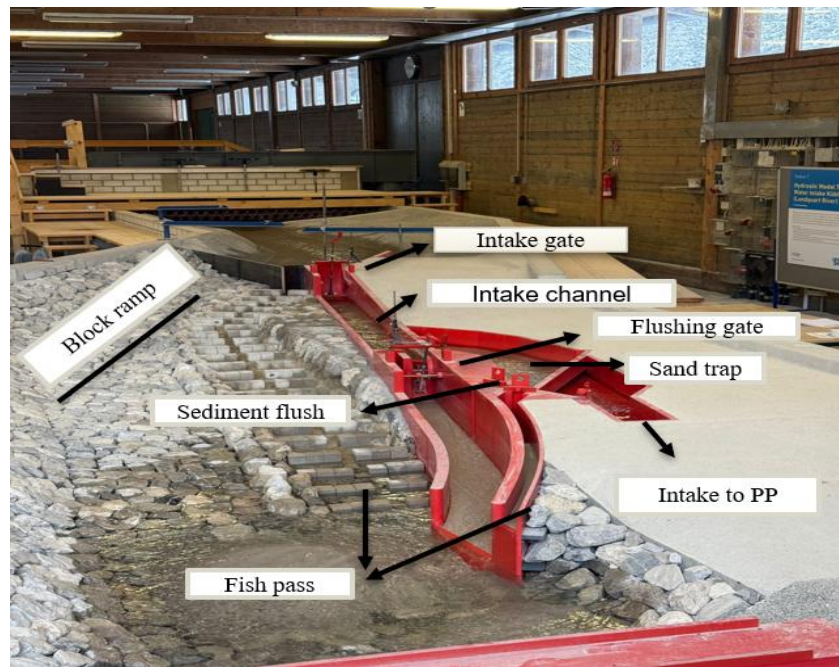


Figure 17 Components of setup.

The block ramp forms the upstream riverbed section and reproduces the bed topography and roughness conditions. While the flushing gate directs water and sediments toward the flushing section, the intake gate controls the flow entering the intake channel. Before the water is routed toward the power plant intake, the sand trap is placed close to the intake to hold onto coarser silt particles. A dedicated fish pass is integrated alongside the main flow path to represent the ecological passage structure.

2.3 EXPERIMENTAL PROCEDURE (Task-1)

In this case, the physical model maintained an upstream discharge of $Q = 12 \text{ m}^3/\text{s}$, which corresponds to a water level of 25.3 cm. The water level was measured at 24.4 cm after the regulating gate was set to permit a discharge of $Q = 9.5 \text{ m}^3/\text{s}$ towards the power plant intake. The left flushing gate stayed closed during the experiment, while the right flushing gate was opened to allow a discharge of $Q = 2.5 \text{ m}^3/\text{s}$.



Figure 18 $Q = 12 \text{ m}^3/\text{s}$, water level 25.3 cm.

Determining the proper flushing gate and regulation gate opening positions under the given flow circumstances was the main purpose of this task. Furthermore, visual observations were made of the flow conditions in the intake channel, flushing section, and nearby river zone.



Figure 19 Upstream WL ($Q = 12 \text{ m}^3/\text{s}$, $h = 40 \text{ cm}$).

The model was updated with a scaled bed load material in order to examine sediment behaviour.

In order to evaluate the features of sediment transport under the specified discharge conditions, the movement, transport path, and deposition of the bed load particles were then observed.

2.3.1 OBSERVATIONS AND RESULTS

The sediment movement was observed for 10 minutes after its introduction into the flow. Fine particles were rapidly transported downstream, while the coarser particles remained within the channel. At the upstream section, the sediment was observed to shift towards the right side of the channel. A flow separation zone was formed near the wall creating a visible gap in the flow region. Some of the fine particles entered the sand trap (Figure 20), whereas noticeable sediment deposition occurred near the flushing gate. Observations and Results



Figure 20 Flow separation near the wall.

Some of the fine particles entered the sand trap (Figure 21), whereas noticeable sediment deposition occurred near the flushing gate.



Figure 21 Sediment deposition in sand trap.

2.4 EXPERIMENTAL PROCEDURE (TASK-2)

An upstream discharge of $Q = 9.5 \text{ m}^3/\text{s}$ was maintained. The regulation gate was opened to allow the entire discharge to flow towards the power plant intake, while both flushing gates remained closed (Figure 22, $Q = 0 \text{ m}^3/\text{s}$).

The objective of this task was to determine the required opening of the regulation gate under the given flow condition with both flushing gates closed. The flow situation was observed, and scaled bed load material was introduced to examine its transport behaviour.



Figure 22 $Q = 9.5 \text{ m}^3/\text{s}$, water level 24.4 cm.

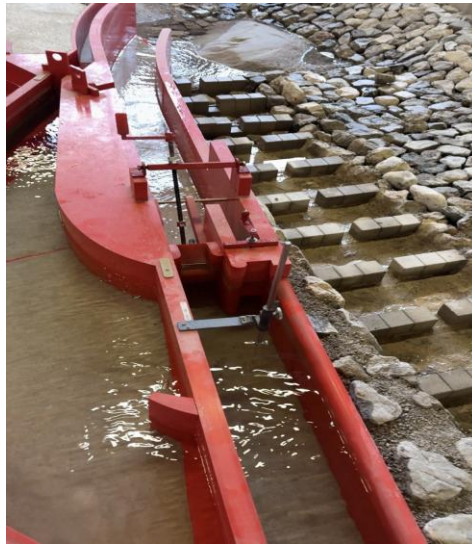


Figure 23 Closed Flush Gates.

2.4.1 OBSERVATIONS AND RESULTS

- The sediment movement was observed for 10 minutes after its introduction into the flow. Fine particles were transported downstream, whereas the coarser particles remained within the channel.
- At the upstream section, the sediment shifted towards the right side, and flow separation created a visible gap near the wall. The sediments moved slowly and were predominantly deposited at the center of the intake channel (Figure 24).



Figure 24 Sediment Deposition in intake channel.

- A significant portion of the particles accumulated near the closed gate, while the remaining particles passed through the open regulation gate.

2.5 EXPERIMENTAL PROCEDURE (TASK-3)

In this task, scaled bed load material was deposited inside the intake channel to represent a worst-case sedimentation state. At first, both flushing gates were fully open, but the incoming gate was maintained closed.

The flushing process was started by applying a discharge of $Q = 2.5 \text{ m}^3/\text{s}$ through the flushing gates. In order to assess the flushing efficiency and its impact on the removal of silt from the intake structure, the intake gate was then opened and the flushing process was monitored.



Figure 25 $Q = 2.5 \text{ m}^3/\text{s}$, water level = 21.8 cm.



Figure 26 Opened Flush gate.

2.5.1 OBSERVATION AND RESULTS

The sediments were initially fully deposited in the upstream section (Figure 27), after which the inlet gates were opened. Increased erosion was observed near the wall (Figure 28) due to the high flow velocity and resulting shear stress.



Figure 27 Intake filled with bedload material.



Figure 28 Initial flow state.

Standing waves (Figure 29) were visible in the intake channel, indicating supercritical flow conditions. After 5 minutes (Figure 30), most of the sediments were flushed out from the intake channel. However, after 8 minutes, a small amount of sediment remained at the upstream center (Figure 31).



Figure 29 Wavy flow at intake channel.



Figure 30 Intake channel after 5 min.



Figure 31 Intake channel after 8 min.

Complete sediment removal was achieved after 9 minutes in the model. According to Froude similarity law, the corresponding prototype time was approximately 34.85 minutes.

$$\frac{T_M}{T_N} = \sqrt{1/15}$$

$$T_N = 9\sqrt{15}$$

2.6 CONCLUSION

The experiment conducted on the 1:15 scaled physical model of the Landsquart River provided a clear understanding of the hydraulic behaviour and sediment transport processes around the intake structure. The results demonstrated that the operation of the regulation and flushing gates significantly influences flow distribution and sediment deposition. Furthermore, the flushing system proved to be effective in removing the accumulated sediments from the intake channel. Overall, the study confirmed the satisfactory hydraulic performance of the intake and flushing arrangement under the tested flow condition.

3.0 BALDENEYSEE

3.1 INTRODUCTION AND SCOPE OF INVESTIGATION

This experiment was carried out at the Obernach hydraulic laboratory to study the flow behaviour of water passing through a sluice gate and stilling basin. The Baldeneysee model helps to understand how high velocity water can be controlled safely using hydraulic structures.

When water passes through a gate opening, the flow area reduces and velocity increases. High velocity flow contains high kinetic energy, which can cause erosion and damage to downstream structures. Therefore, a stilling basin is provided to reduce the flow energy by forming a hydraulic jump.

The aim of this experiment is to understand how the flow changes for different gate openings and how the guiding extension influences the hydraulic jump and energy dissipation.

The main objectives of this experiment are :

- Determine the discharge capacity for different gate opening.
- Analyse flow conditions upstream, inside stilling basin and downstream.
- Observe hydraulic jump formation
- Study energy dissipation process
- Evaluate influence of guiding extension

This experiment helps to understand practical hydraulic engineering design and behaviour of open channel flow.

3.2 THEORETICAL BACKGROUND

When water flows through a sluice gate, the opening controls the discharge. As the opening becomes larger, more water passes through the gate and the discharge increases. When water passes through the gate, velocity increases and flow becomes supercritical. Supercritical flow has high velocity and low depth. This flow cannot remain stable for long distance, therefore the flow suddenly changes to subcritical flow inside the stilling basin.

The sudden transition from supercritical flow to subcritical flow is called hydraulic jump.

Hydraulic jump is important because:

- it reduces flow velocity
- it dissipates energy

- it prevents erosion
- it protects downstream structures

Guiding extension is used to guide the water jet smoothly into the stilling basin. It helps to stabilise the hydraulic jump and improves energy dissipation efficiency.

3.3 MODEL SETUP AND INSTRUMENTATION

A Froude-scaled physical model at a scale of 1:25 of the Baldeneysee sluice gate and stilling basin was constructed.

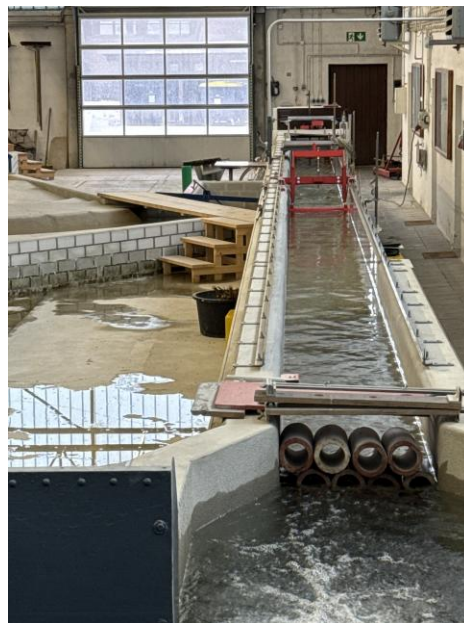


Figure 32 Baldeneysee River Model.

MODEL PARAMETERS :

PARAMETER	VALUE
Scale	1:25
Channel width	0.50 m
Weir crest level	45.40 m asl
Upstream gauge	45.10 m asl
Downstream gauge	42.75 m asl

Table 3 Given Parameters.

The discharge was measured using flow measurement instruments and water levels were measured using point gauges. The stilling basin was used to observe hydraulic jump formation and energy dissipation behaviour.

3.4 EXPERIMENTAL PROCEDURE AND OBSERVATIONS

The experiment was performed for different gate openings to analyse discharge capacity and flow behaviour.

3.4.1 CASE 1 – GATE OPENING 0.25 M (5°)

Determination of discharge capacity

The measured discharge for the model is: $Q_m = 8.1 \text{ l/s}$

Using scaling relation: $Q_P = Q_M * L^{2.5}$

Prototype discharge obtained: $Q_p = 25.31 \text{ m}^3/\text{s}$



Figure 33 Upstream Water Level.



Figure 34 Downstream Water level.



Figure 35 Observed Discharge.

Flow behaviour

Upstream side

- Flow appears calm and smooth before reaching the gate.
- Flow contracts near the gate opening.
- Velocity increases as water approaches the gate.
- Flow becomes supercritical immediately after passing the gate.

Stilling basin

- High velocity jet enters the stilling basin.
- Hydraulic jump forms clearly inside the stilling basin.
- Strong turbulence and roller formation observed.

Large amount of energy dissipation occurs inside the stilling basin.

Downstream side

- Flow becomes subcritical after hydraulic jump.
- Water surface becomes more stable and uniform.
- Turbulence reduces gradually further downstream.

3.4.2 CASE 2 – GATE OPENING 1.48 M (25°) - WITHOUT GUIDING EXTENSION

Determination of discharge capacity

Measured discharge for model: $Q_m = 38.3 \text{ l/s}$

Using scaling relation: $Q_P = Q_M * L^{2.5}$

Prototype discharge: $Q_p = 119.68 \text{ m}^3/\text{s}$



Figure 36 Upstream Water Level.

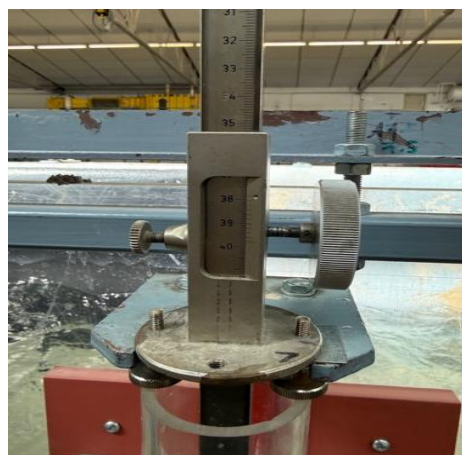


Figure 37 Downstream Water Level.

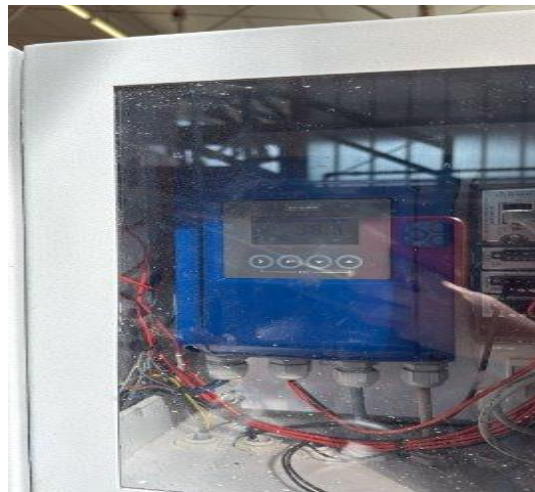


Figure 38 Observed Discharge.

Behaviour

Upstream side

- Flow velocity increases significantly due to larger gate opening.
- Flow spreads more freely because there is no guiding extension.
- Small vortices form near upstream wall due to uneven flow direction.

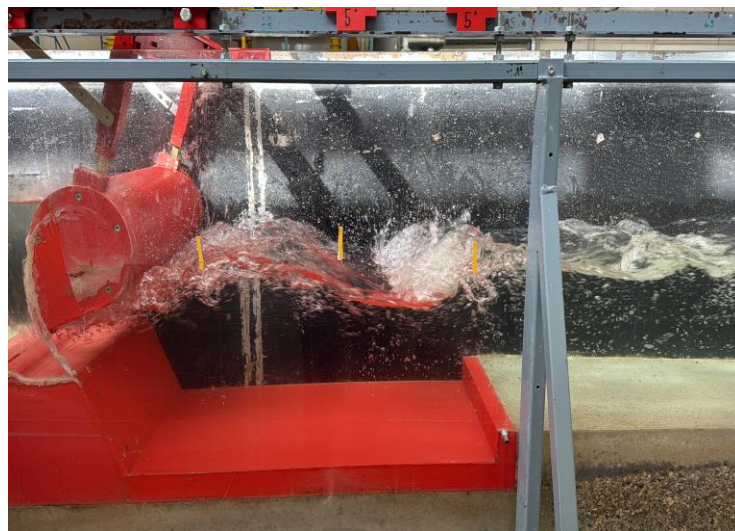


Figure 39 Flow Situation observed without guiding extension.

Stilling basin

Strong hydraulic jump forms inside the stilling basin.

- Hydraulic jump is slightly shifted downstream because flow is not guided.
- Large roller and recirculation zone observed.

- Turbulence is very high and spread over a larger area.
- Energy dissipation occurs in the stilling basin but is less controlled.
- Flow fluctuations are higher compared to guided condition.

Downstream side

- Flow changes from supercritical to subcritical after the jump.
- Water surface becomes stable after some distance.
- Some turbulence continues downstream.

3.4.3 CASE 3 – GATE OPENING 1.48 M (25°) - WITH GUIDING EXTENSION

Flow behaviour

Upstream side

- Guiding extension directs the water smoothly into the stilling basin.
- Flow becomes more aligned and uniform.
- Jet follows smoother path after passing the gate.



Figure 40 Flow Situation observed with guiding extension.

Stilling basin

- Hydraulic jump forms closer to the basin floor.
- Jump position becomes more stable and controlled.
- Turbulence is present but more uniform.
- Energy dissipation becomes more efficient and concentrated.

- Flow oscillations reduce compared to case without guiding extension.
- Roller formation is smaller but more stable.

Downstream side

- Flow clearly changes from supercritical to subcritical.
- Water surface becomes smoother and more uniform.
- Less turbulence observed compared to case without guiding extension.

3.5.RESULTS

OBJECTIVE	RESULTS
Determine discharge capacity	Discharge increases with increase in gate opening
Analyse flow conditions	Flow changes from subcritical to supercritical and again to subcritical
Observe hydraulic jump	Hydraulic jump clearly forms in stilling basin
Observe energy dissipation	Maximum energy dissipation occurs in stilling basin
Influence of guiding extension	Guiding extension improves flow stability

Table 4 Result Obtained

3.6.CONCLUSION

The Baldeneysee experiment shows how gate opening affects discharge capacity and flow behaviour. For smaller gate openings, flow velocity is lower and hydraulic jump is weaker. For larger gate openings, flow

velocity increases and strong hydraulic jump forms, which produces high turbulence and high energy dissipation.

Without guiding extension, the hydraulic jump is strong but less controlled. Turbulence spreads over a larger area and energy dissipation is less uniform. With guiding extension, flow becomes smoother and hydraulic jump becomes more stable. Energy dissipation becomes more efficient and controlled. Therefore, guiding extension improves hydraulic performance of stilling basin and reduces risk of erosion downstream. The experiment helps to understand real hydraulic behaviour and importance of proper design of hydraulic structures.